Retail Sales Performance Analysis

An evidence-led review of sales, profitability and commercial performance

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1 Understanding what drives sales and profit

A practical statistical review of 700 retail transactions, examining where revenue is generated, which parts of the business are most profitable, and how discounting relates to financial performance.

Analysis focus: descriptive analytics · segment and product comparisons · discount analysis · statistical inference · regression modelling

ⓘ Note

About this edition

This report has been rewritten for a business audience while retaining the statistical findings from the analysis. It uses the verified summary results already produced from the cleaned dataset. The charts below are built from those reported aggregate results; they do not invent transaction-level observations or unreported monthly values.

1.1 Executive summary

The analysis covers **700 cleaned sales records** across five business segments, five countries and six products. The overall picture is positive: the business generated substantial sales and positive profit. However, the results are not evenly distributed. Government accounts contribute the largest share of revenue and total profit, while Enterprise is the clearest profitability concern.

Discounting deserves careful attention. Higher discount categories are associated with lower average profit and lower profit margins. The statistical tests also indicate that at least some differences in profit across discount categories are unlikely to be due to random variation alone. This does not prove that discounts cause lower profit, because customer type, product mix and deal size may also influence the results.

The regression model explains about **30.2%** of the observed variation in profit. Units sold are positively associated with profit, while discount rate is negatively associated with profit after accounting for the included business categories. The model is useful for identifying patterns, but it is not accurate enough to explain every difference in transaction profitability.

Cleaned records

700

After removing blank and duplicate rows

Average sales

169,609

Per transaction (reported mean)

Average profit

24,134

Per transaction (reported mean)

Model R-squared

30.2%

Variation in profit explained

1.2 1. Purpose and business questions

Sales figures alone do not tell the whole story. A business can generate high revenue and still struggle to make a healthy profit if its costs, pricing or discounts are not well managed. This analysis therefore looks beyond total sales to understand the quality of the revenue being generated.

The report addresses five practical questions:

1. Which business segments contribute most to sales and profit?
2. How does performance differ across countries and products?
3. Are higher discounts associated with weaker profitability?
4. How did sales and profit move over the observed reporting period?
5. Which factors are statistically associated with transaction-level profit?

The work is observational. It identifies patterns and relationships in the available records; it does not establish that one business decision directly caused another outcome.

1.3 2. Data and preparation

The original file contained **710 rows and 15 columns**. Initial preparation identified five blank rows and five duplicate records. After removing these records and creating additional analysis variables, the working dataset contained **700 observations and 18 variables**.

The cleaned data covered five business segments:

- Government
- Small Business
- Channel Partners
- Midmarket
- Enterprise

It also covered five countries—Canada, France, Germany, Mexico and the United States—and six products: Amarilla, Carretera, Montana, Paseo, Velo and VTT.

1.3.1 Data-quality checks

The checks found no missing values in the cleaned dataset. The accounting identity `Profit = Sales - COGS` held within a tolerance of 0.001. Gross Sales exceeded Sales in 647 records and equalled Sales in 53 records; Sales never exceeded Gross Sales.

These checks provide reassurance that the prepared data are internally consistent on the fields examined. They do not rule out every possible source-data or business-definition issue.

1.4 3. Overall financial picture

The financial measures show a wide gap between the typical transaction and the largest transactions. For example, mean sales were approximately 169,609, while median sales were 35,540. This difference indicates a strongly right-skewed distribution: a relatively small number of large transactions pull the average upward.

Profit shows a similar pattern. Mean profit was approximately 24,134, compared with a median of 9,242. The mean is therefore not representative of every individual transaction, and the median provides useful additional context.

1.4.1 Financial summary

Summary of the main financial measures

Measure	Mean	Median	Minimum	Maximum
Sales	169,609.07	35,540.20	1,655.08	1,159,200.00
Gross Sales	182,759.43	37,980.00	1,799.00	1,207,500.00
Cost of Goods Sold	145,475.21	22,506.25	918.00	950,625.00
Profit	24,133.86	9,242.20	-40,617.50	262,200.00
Discount Amount	13,150.35	2,585.25	0.00	149,677.50

1.4.2 Profit outliers: unusual does not automatically mean wrong

The interquartile-range rule identified 103 observations (14.71%) outside the calculated profit bounds. Most were unusually high-profit records: 97 were above the upper bound and six were below the lower bound.

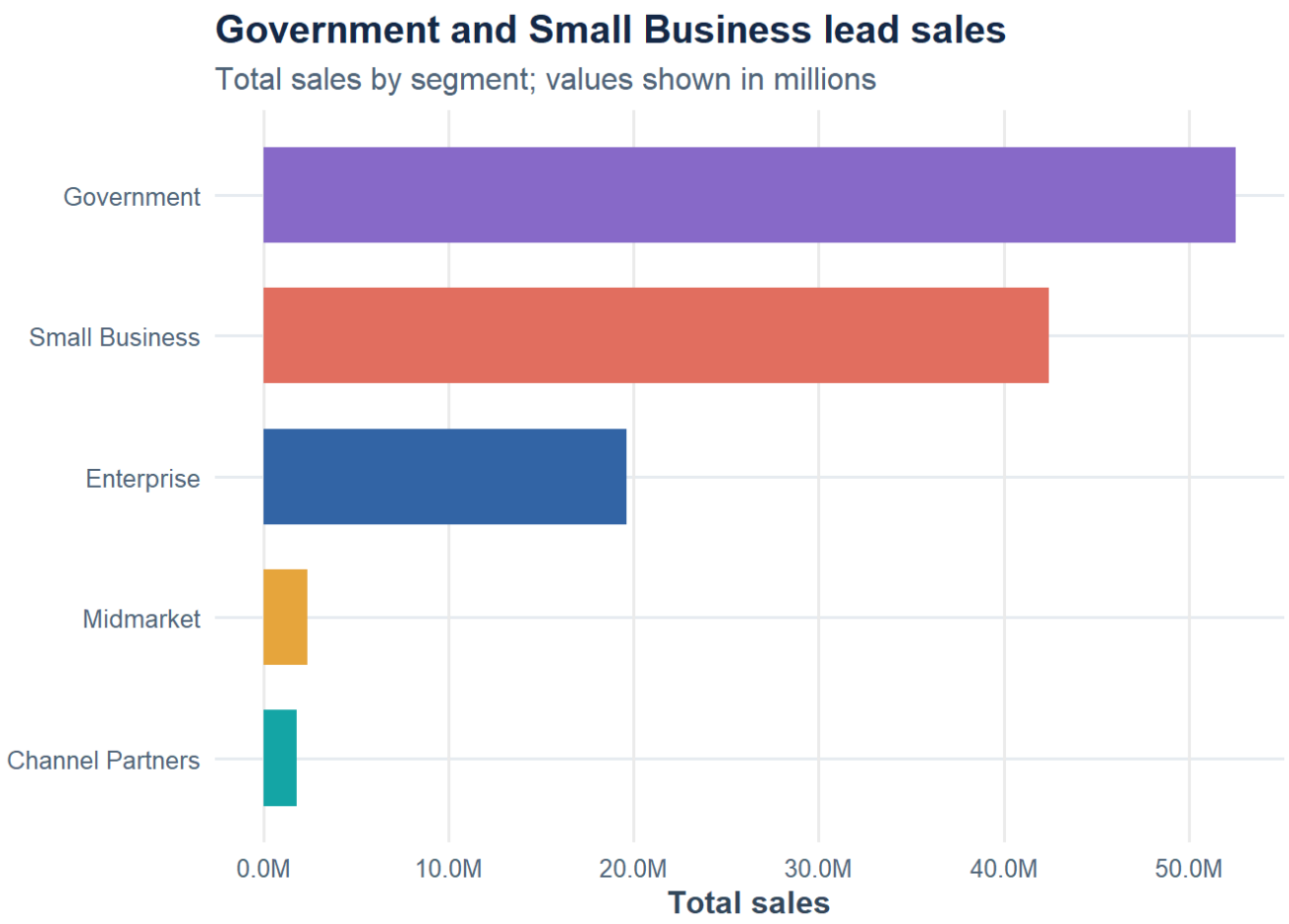
These observations were retained. In a commercial dataset, unusually large transactions may represent genuine high-value deals rather than data errors. Removing them without a clear business reason could hide an important part of the company's performance.

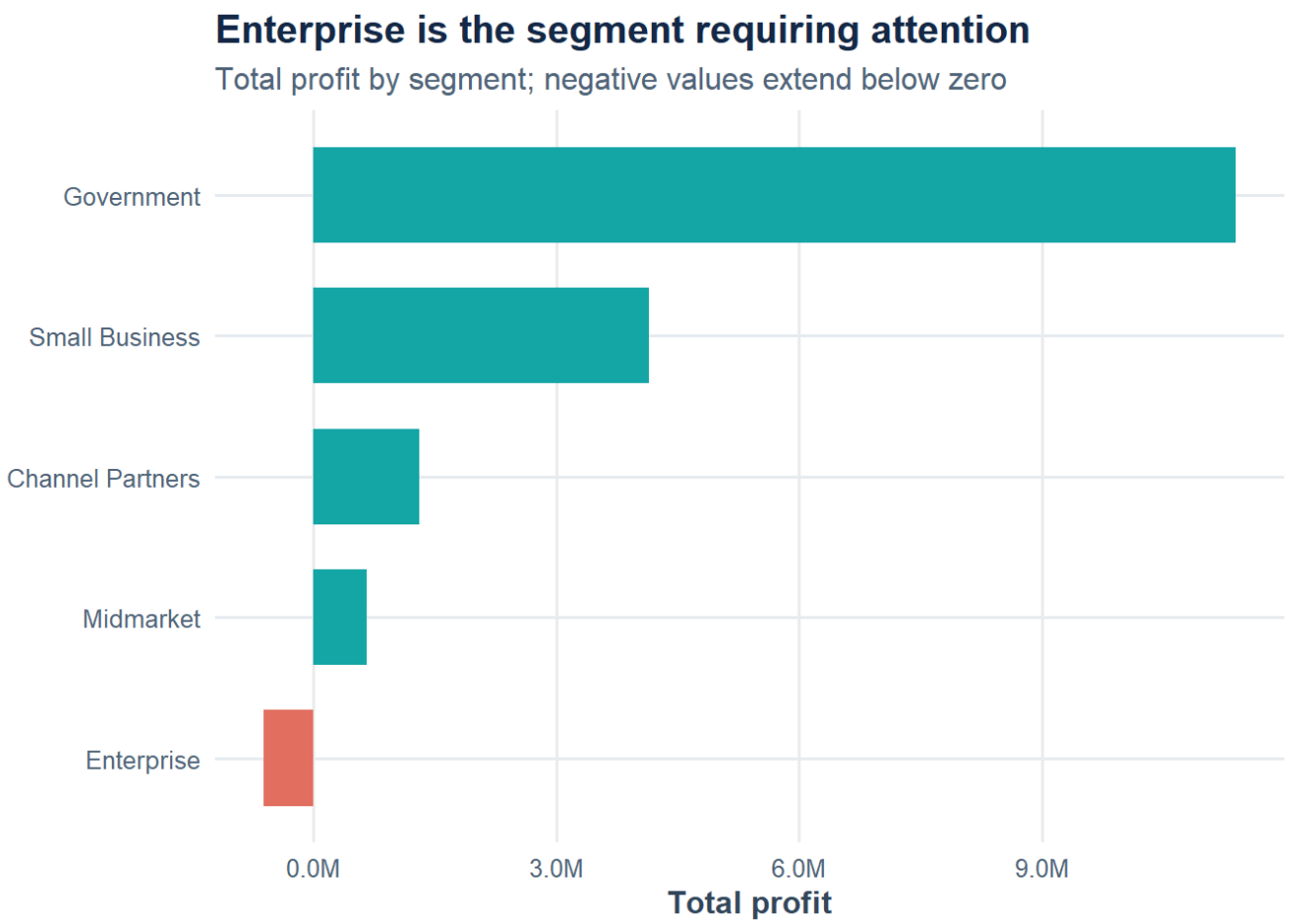
1.5 4. Performance across business segments

The segment comparison reveals an uneven distribution of commercial performance. Government is the largest segment by both sales and total profit. Channel Partners, although much smaller in sales volume, records the highest profit margin. Enterprise is the only segment with a negative total profit and a negative margin.

Sales and profitability by business segment

Segment	N	Sales	Profit	Avg_Profit	Margin
Government	300	52,504,261	11,388,173	37,961	21.69%
Small Business	100	42,427,919	4,143,168	41,432	9.77%
Channel Partners	100	1,800,594	1,316,803	13,168	73.13%
Midmarket	100	2,381,883	660,103	6,601	27.71%
Enterprise	100	19,611,694	-614,546	-6,145	-3.13%


Government and Small Business lead sales
Total sales by segment; values shown in millions
Government
Small Business
Enterprise
Midmarket
Channel Partners
0.0M
10.0M
20.0M
30.0M
40.0M
50.0M
Total sales



What this means for the business. Government is the main contributor to the current sales and profit totals, so maintaining this relationship is important. Channel Partners has a striking reported margin of 73.13%, but its sales base is comparatively small. It may be worth investigating whether this model can be expanded without changing its cost structure.

Enterprise requires a different response. Its reported total profit is negative, despite sales of almost 19.6 million. Management should examine the cost-to-serve, pricing, contract terms and discount arrangements for this segment before pursuing further revenue growth there.

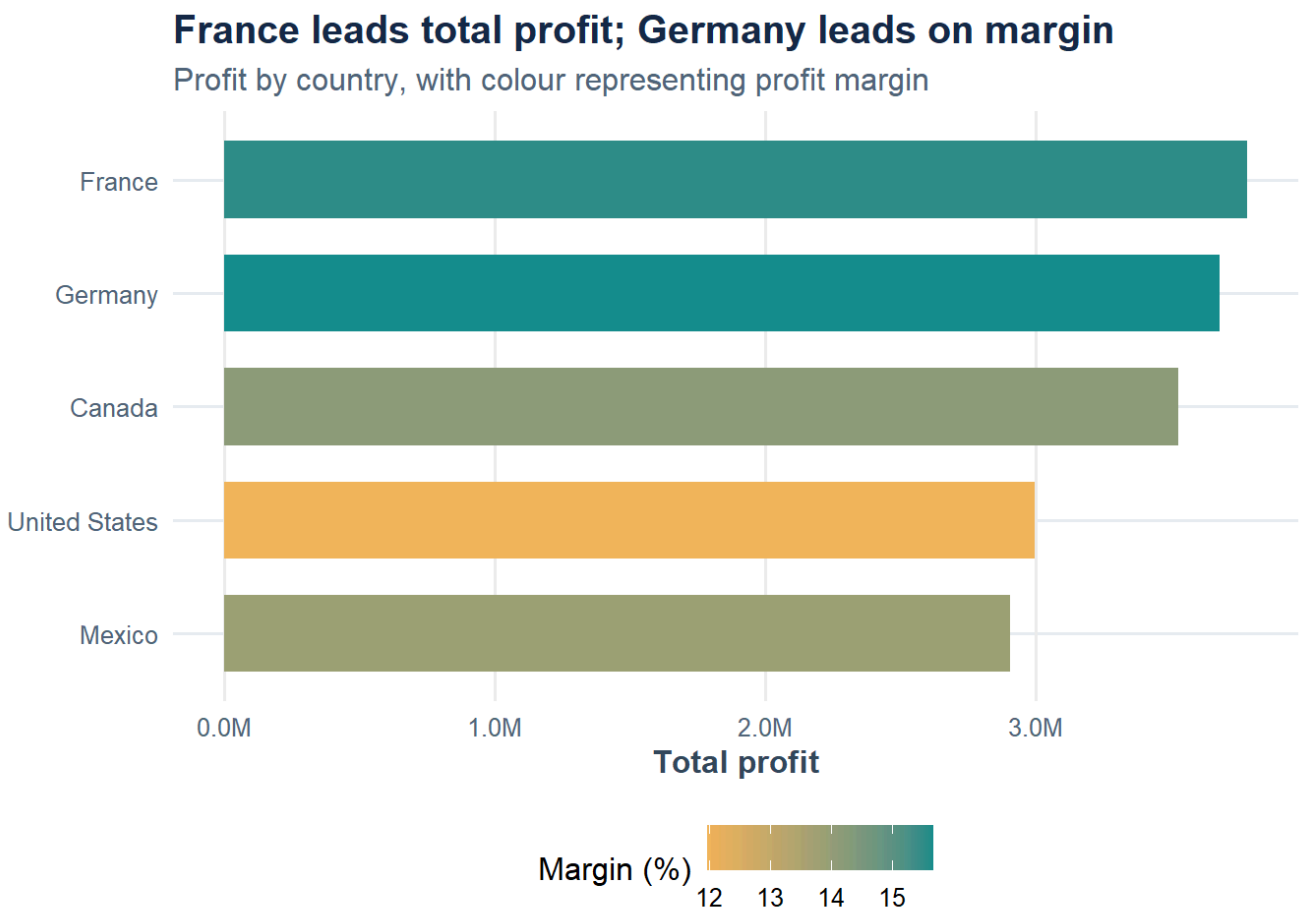
1.6 5. Country performance

Country-level results are more balanced than the segment results. The United States has the highest reported sales total, while France has the highest total profit. Germany records the highest profit margin, narrowly ahead of France.

Country-level financial performance

Country	Sales	Profit	Avg_Profit	Margin

Country	Sales	Profit	Avg_Profit	Margin
France	24,354,172	3,781,021	27,007	15.53%
Germany	23,505,341	3,680,389	26,288	15.66%
Canada	24,887,655	3,529,229	25,209	14.18%
United States	25,029,830	2,995,541	21,397	11.97%
Mexico	20,949,352	2,907,523	20,768	13.88%



The difference between the sales leader and profit leader is useful. It shows why revenue should not be used as the only measure of market performance. The United States generated the largest sales total, but its reported margin was the lowest of the five countries. France and Germany converted their sales into profit more effectively.

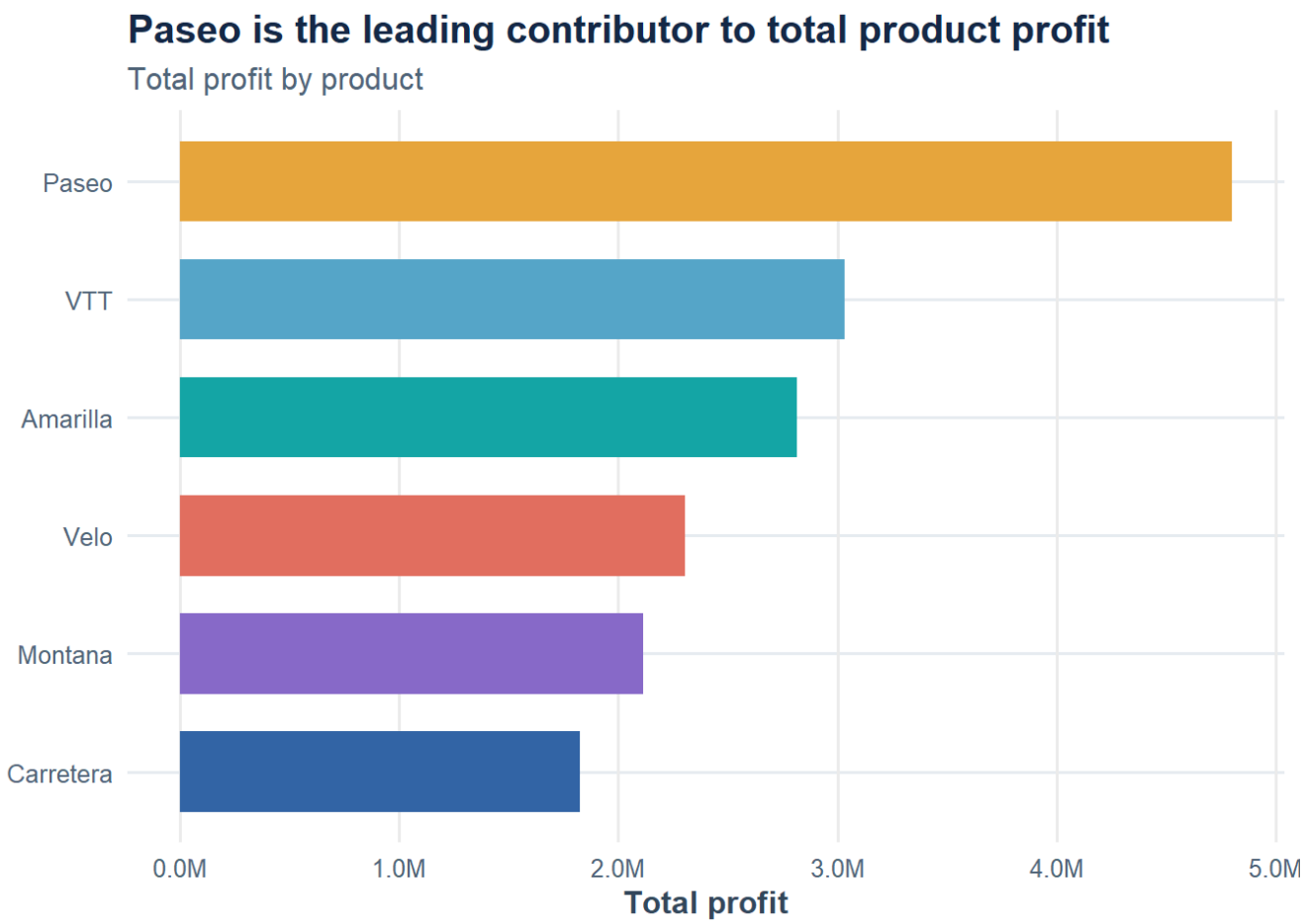
1.7 6. Product performance

Paseo contributed the highest total sales and profit. Amarilla recorded the highest average profit per observation and the highest profit margin among the six products. This distinction matters: a product

can lead in total contribution because it appears more frequently, while another can perform better on a per-record basis.

Product-level financial performance

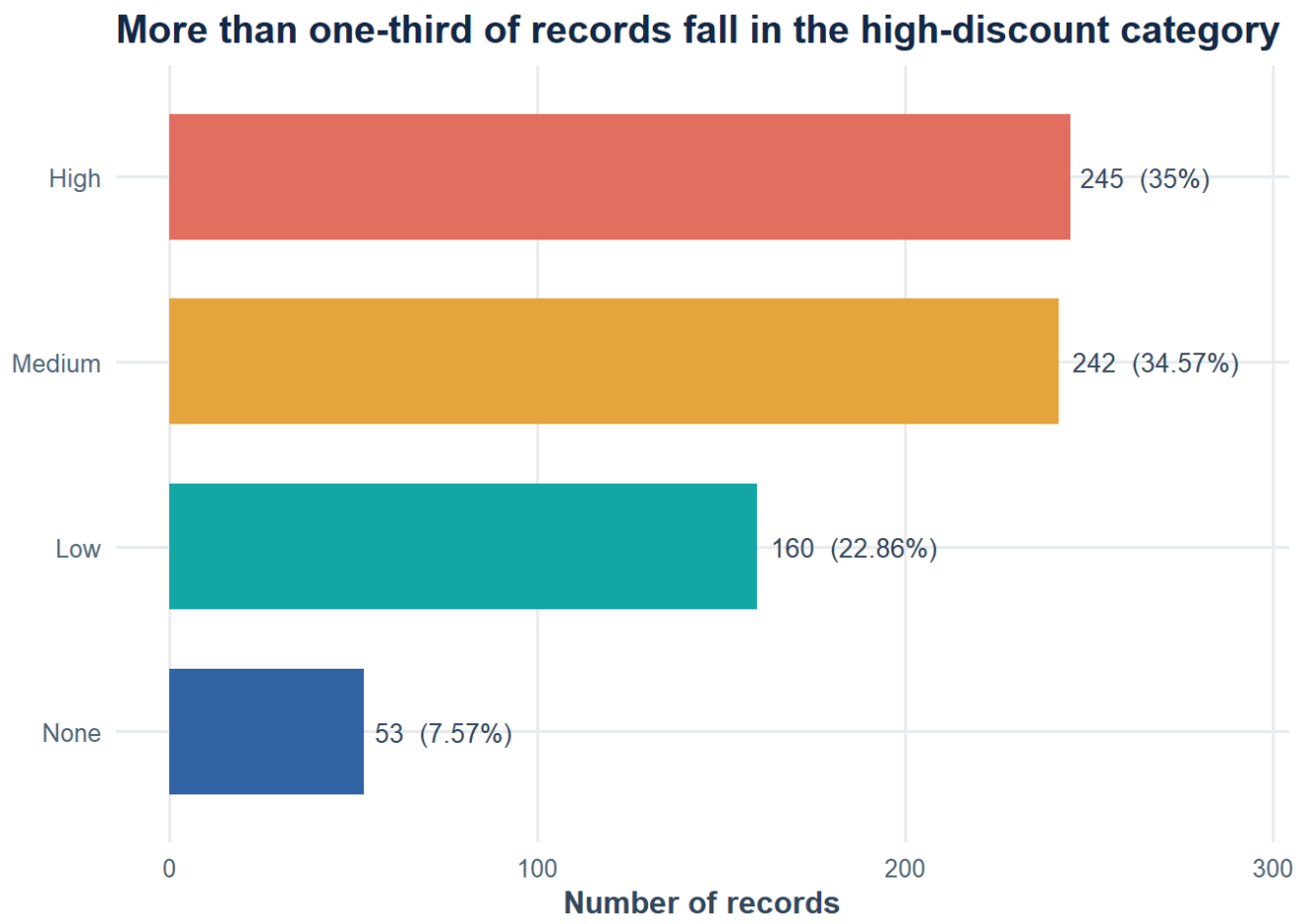
Product	Sales	Profit	N	Avg_Profit	Margin
Paseo	33,011,144	4,797,438	202	23,750	14.53%
VTT	20,511,921	3,034,608	109	27,840	14.79%
Amarilla	17,747,116	2,814,104	94	29,937	15.86%
Velo	18,250,059	2,305,992	109	21,156	12.64%
Montana	15,390,802	2,114,755	93	22,739	13.74%
Carretera	13,815,308	1,826,805	93	19,643	13.22%



A sensible product strategy would protect Paseo's contribution while examining what makes Amarilla's per-observation profitability stronger. Carretera and Montana may also deserve a closer review of pricing, product costs and sales mix.

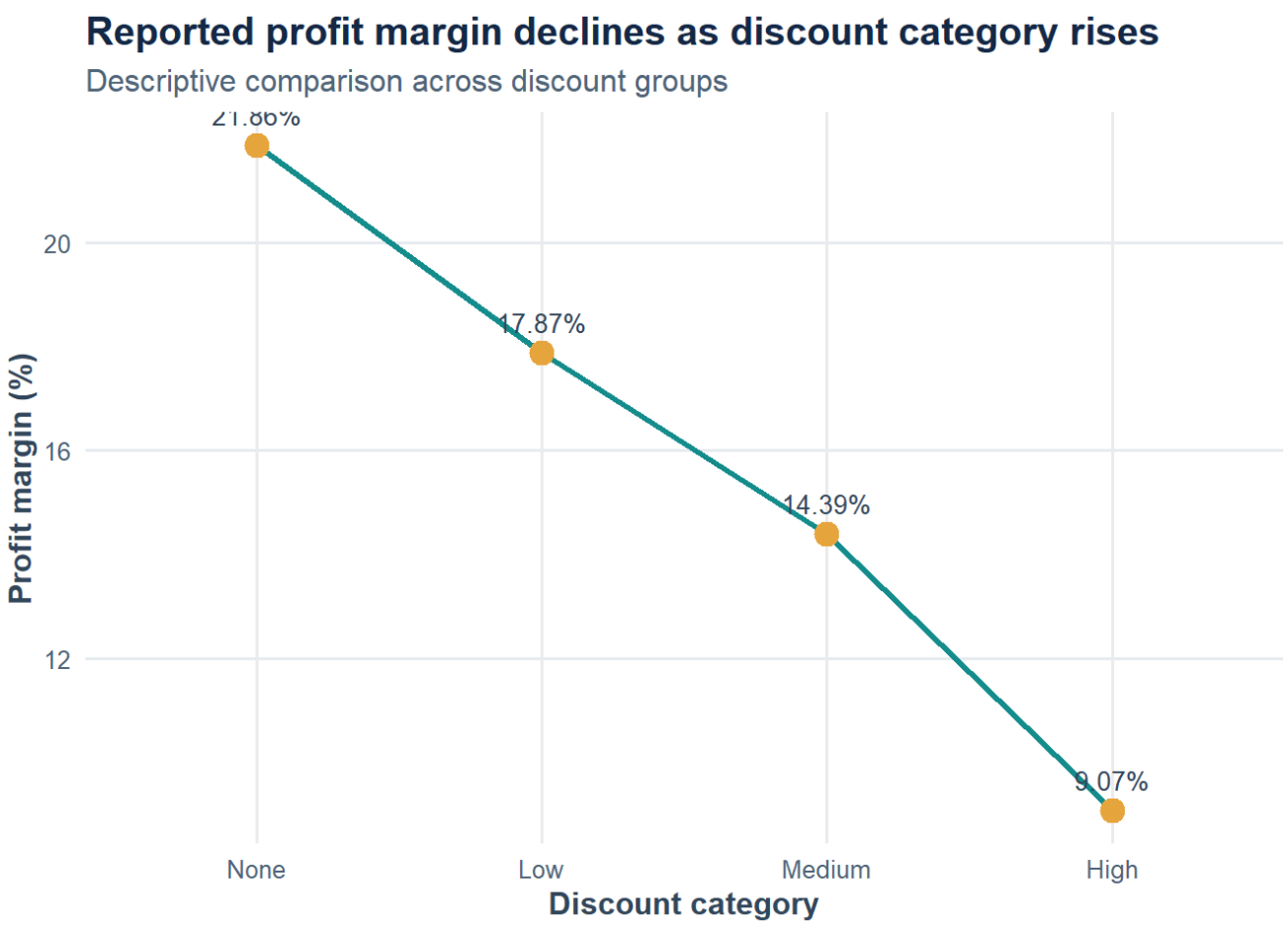
1.8 7. Discounts and profitability

Discounting is common across the records, but the distribution is concentrated in the higher categories. High discounts account for 245 records (35.00%), while only 53 records (7.57%) have no discount.



Profitability indicators by discount category

Category	Avg_Sales	Avg_Profit	Rate	Total_Profit	N	Margin
None	149,880	32,763	0%	1,736,455	53	21.86%
Low	216,436	38,680	2.38%	6,188,858	160	17.87%
Medium	160,250	23,056	6.97%	5,579,523	242	14.39%
High	152,541	13,832	12.51%	3,388,867	245	9.07%



The pattern is clear in the aggregate results. Average profit falls from 38,680 in the low-discount category to 13,832 in the high-discount category. Profit margin also declines from 17.87% to 9.07%.

The no-discount group has a margin of 21.86%, but it contains only 53 records. The high-discount group contains 245 records. These groups may differ in customer type, product mix, deal size and other characteristics, so the comparison should not be interpreted as a controlled experiment.

1.8.1 Statistical evidence

A Kruskal–Wallis test found a statistically significant difference in profit across discount categories, (H(3)=18.53), (p=0.000342). The estimated effect size was small ((^2=0.0223)). The result suggests that the distributions are not all alike, but the magnitude of the overall difference is modest.

The Dunn post-hoc comparisons with Holm adjustment found statistically significant differences between High and Low discounts ((p=0.002611)), High and Medium ((p=0.022610)), and High and None ((p=0.007700)). The remaining pairwise comparisons were not statistically significant at the 5% level.

Some segment-by-discount combinations contain very few observations. In particular, the Small Business no-discount group has four records and the Enterprise no-discount group has six. Their

averages should therefore be treated cautiously.

1.9 8. Performance over time

The reporting period runs from September 2018 to December 2019. The available summary identifies October 2019 as the highest-sales month, with sales of 12,375,820, and December 2019 as the highest-profit month, with profit of 2,025,765.90.

The movement around the end of 2019 was pronounced. From October to November 2019, sales fell by 56.49% and profit fell by 66.07%. From November to December, sales increased by 122.85% and profit increased by 235.06%.

Profit margin also varied over the period. October 2018 recorded the highest reported monthly margin at 17.83%, while November 2018 recorded the lowest at 10.53%. The margin was 11.23% in November 2019 and recovered to 16.88% in December 2019.

A full month-by-month line chart is intentionally not drawn here because the complete 16 monthly values are not included in the summary results used for this report. This avoids presenting invented intermediate values as if they were observed.

1.9.1 Annual comparison: an important qualification

Year	Sales	Profit	Average profit per record	Records	Profit margin
2018	26,415,256	3,878,465	22,162.65	175	14.68%
2019	92,311,095	13,015,238	24,790.93	525	14.10%

The reported totals increased substantially in 2019. However, the dataset contains only four months of 2018 and twelve months of 2019. The reported growth rates—249.46% for sales and 235.58% for profit—are therefore not like-for-like annual growth rates. They mainly reflect the unequal number of months represented.

1.10 9. Relationships between sales, discounts and profit

Spearman correlation was used to assess monotonic relationships between selected financial variables. This approach is useful where the data may be skewed or influenced by unusually large transactions.

Variable pair	Spearman correlation	Interpretation
Units Sold and Sales	0.376	Positive association
Units Sold and Discount	-0.017	Negligible association
Units Sold and Profit	0.295	Positive association
Sales and Discount	-0.073	Very weak negative association
Sales and Profit	0.581	Moderate positive association
Discount and Profit	-0.175	Weak negative association

The strongest reported relationship is between Sales and Profit (()). Transactions with higher sales tend to have higher profit, although the relationship is far from perfect. Discount and Profit have a weak negative association (()), consistent with the descriptive discount comparisons.

The correlation between Sales and Discount was weak and not statistically significant at the 5% level ((p=0.05434)). Correlation describes association; it does not show that changing one variable will necessarily change another.

1.11 10. Regression modelling

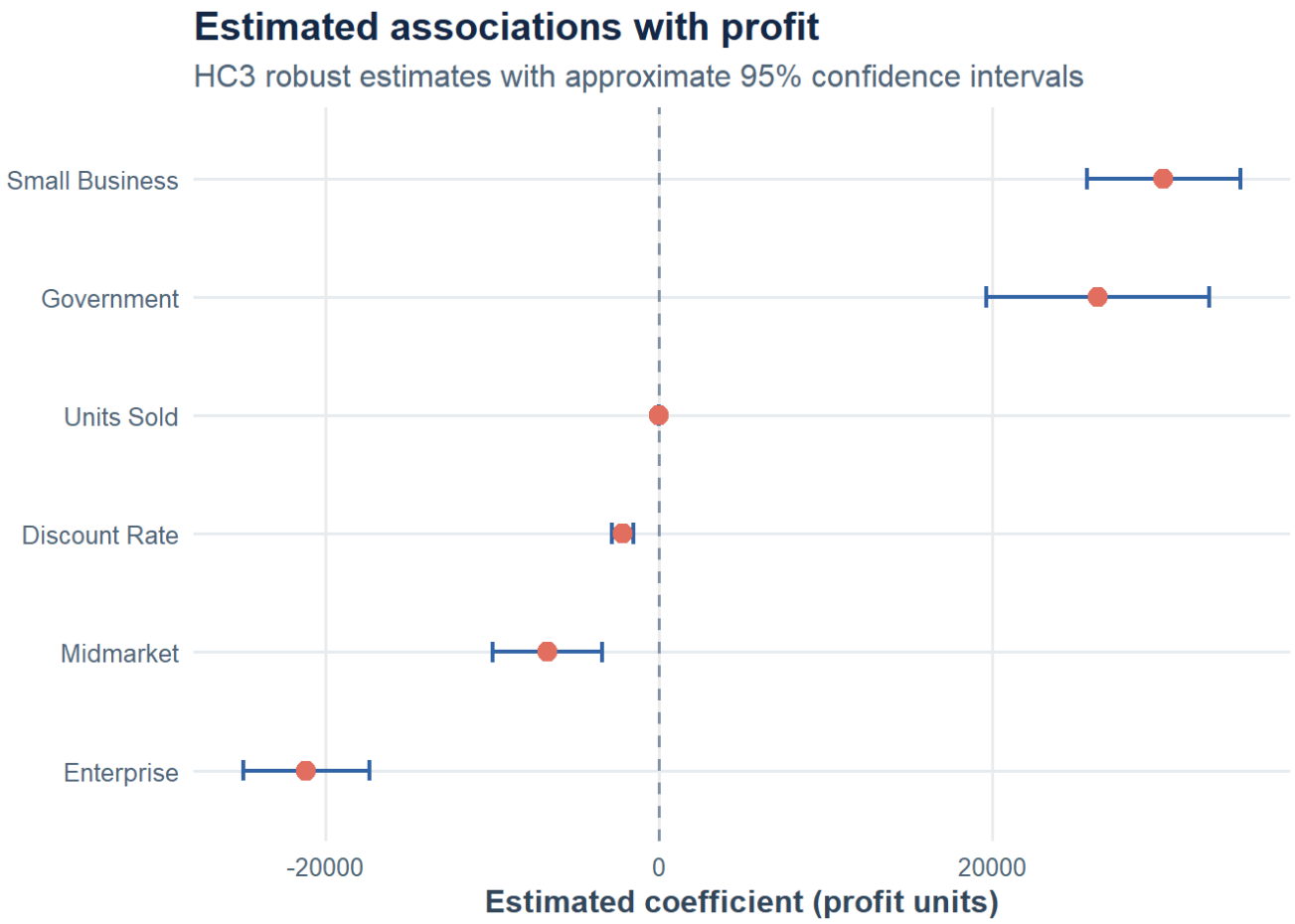
A multiple linear regression model was fitted to examine how Units Sold, Discount Rate, Segment, Product and Country relate to transaction-level profit.

The model used Channel Partners, Amarilla and Canada as the reference categories. The overall model was statistically significant, (F(15,684)=19.71), (p<0.001). Its (R^2) was 0.3018 and adjusted (R^2) was 0.2864. In practical terms, the included predictors account for about 30% of the variation in profit in this dataset. A substantial share remains unexplained.

1.11.1 Main coefficient results

Predictor	Estimate	HC3 robust SE	Robust p-value
Intercept	13,594.96	5,944.45	0.0225
Units Sold	12.80	1.99	<0.001
Discount Rate	-2,179.37	323.21	<0.001
Enterprise	-21,153.60	1,932.88	<0.001
Government	26,343.35	3,422.40	<0.001

Predictor	Estimate	HC3 robust SE	Robust p-value
Midmarket	-6,680.12	1,679.28	<0.001
Small Business	30,301.73	2,341.03	<0.001



Units Sold has a positive coefficient: holding the other included predictors constant, one additional unit sold is associated with an estimated 12.80-unit increase in profit. Discount Rate has a negative coefficient of -2,179.37, indicating that higher discount rates are associated with lower profit after adjustment for the included variables.

Compared with Channel Partners, Government and Small Business have positive estimated profit differences, while Enterprise and Midmarket have negative differences. These comparisons are conditional on the model specification and the selected reference categories; they should not be interpreted as causal effects.

1.11.2 Why robust standard errors were used

The Breusch–Pagan test indicated heteroscedasticity ((^2=164.68), (df=15), (p<0.001)). This means the residual variance is not constant across the fitted values or predictor combinations. HC3

heteroscedasticity-consistent standard errors were therefore used for the main coefficient interpretation.

The Midmarket coefficient illustrates why this matters. Under conventional standard errors its p-value was 0.193, whereas under HC3 robust standard errors it was statistically significant ((p<0.001)). The coefficient itself did not change; the estimated uncertainty around it did.

The variance inflation factors were all close to one (approximately 1.003 to 1.019), providing no evidence of problematic multicollinearity among the included predictors.

Influence checks identified 34 observations (4.86%) above the Cook's distance screening threshold of (4/n=0.0057). The largest Cook's distance was 0.0378. Thirteen observations (1.86%) had absolute standardised residuals above three, mostly high-profit Government records. These cases warrant attention, but a screening threshold alone is not a reason to delete observations.

Residual plots showed departures from normality in the tails and evidence of non-constant variance. The model should therefore be treated as a useful explanatory summary rather than a complete description of every transaction.

1.12 11. Key findings

The analysis points to six main conclusions:

1. **Revenue is concentrated.** Government and Small Business account for the largest sales totals, with Government also contributing the largest total profit.
2. **Enterprise profitability is a concern.** It is the only segment with negative total profit and a negative reported margin.

. **Channel Partners has a strong margin but a smaller base.** Its reported 73.13% margin stands out, although its total sales are much lower than those of the larger segments. 4. **Discounting is associated with weaker margins.** The high-discount category has the lowest reported average profit and margin, and the inferential tests identify several significant group differences. 5. **Product and country leaders differ depending on the measure.** Paseo leads total product profit, Amarilla leads average profit per record and margin, France leads country profit, and Germany leads country margin. 6. **The regression model identifies meaningful associations, not a full explanation.** Units Sold and selected segment contrasts are positively associated with profit, while Discount Rate and Enterprise are negatively associated with it. The model explains about 30.2% of profit variation.

1.13 12. Business recommendations

1.13.1 Review Enterprise contracts and costs

Enterprise should be the first priority for a focused profitability review. Management should examine whether contract pricing covers delivery and service costs, whether discounts are appropriate for the value of each account, and whether some contracts are consistently loss-making.

1.13.2 Protect profitable Government relationships

Government is the largest contributor to sales and total profit. Maintaining service quality and contract discipline in this segment is important. The aim should be to protect profitable revenue rather than pursue volume at any price.

1.13.3 Understand and scale the Channel Partners model

The reported Channel Partners margin is unusually high relative to the other segments. Before expanding this model, management should verify the underlying cost allocation and determine whether the same economics can be maintained at a larger scale.

1.13.4 Make discount decisions more selective

The results suggest that high discounts deserve closer scrutiny. A practical next step is to set minimum margin thresholds, require approval for exceptional discounts, and assess discounts by customer segment and product rather than applying a single rule across the business.

1.13.5 Use product-level performance to guide commercial planning

Paseo is an important contributor to total profit, while Amarilla performs well on average profit and margin. These products should be reviewed for opportunities to protect availability, pricing and customer demand. Lower-performing products should be assessed using both margin and strategic importance.

1.13.6 Monitor profit, not just sales

Management reporting should show sales, profit, margin, discount rate and transaction volume together. This reduces the risk of celebrating revenue growth that does not translate into healthy profitability.

1.14 13. Limitations and interpretation

The analysis has several limitations. First, it is based on observational sales records, so the relationships identified should not be treated as causal. Second, the regression model explains a moderate proportion of profit variation, leaving many transaction-level influences outside the model.

Third, heteroscedasticity and influential observations mean that model results require cautious interpretation, even with robust standard errors.

The annual comparison is also affected by unequal reporting periods: four months are represented in 2018 and twelve in 2019. The growth percentages should not be presented as directly comparable year-on-year growth.

Finally, some segment-by-discount groups are small. Their averages can change substantially when only a few records are added or removed. Decisions should therefore consider group size and business context alongside the reported percentages.

1.15 14. Conclusion

This analysis shows that the business is generating substantial sales and positive overall profit, but performance varies considerably across segments, products, countries and discount categories. Government is the strongest contributor to total profit, while Enterprise stands out as a segment where revenue is not translating into positive returns. Paseo leads total product profit, whereas Amarilla has the strongest average profit and margin.

Discounting is an important management issue. Higher discount categories are associated with lower average profit and lower margins, and the statistical tests support the presence of differences between selected discount groups. The regression results reinforce the negative association between discount rate and profit after accounting for the included business characteristics.

The most useful next step is not simply to increase sales. It is to improve the quality of sales: protect profitable accounts, review loss-making contracts, understand why some segments achieve stronger margins, and make discount decisions using clear profitability thresholds.

The findings provide a practical starting point for better commercial monitoring. They should be revisited as more transaction data become available and as the business incorporates additional information about customer acquisition, operating costs, contract terms and product strategy.

1.16 Appendix A: Reproducibility

The analysis was conducted using R and Quarto. The report presents the summary results and model outputs from the cleaned 700-record dataset. The charts are generated from the reported aggregate tables. Transaction-level distribution plots, complete monthly trend charts and diagnostic plots should be regenerated from the underlying cleaned records when the report is rendered alongside the source dataset.